

IPOInsight – AI-Powered IPO Investment Decision Support System

Abstract

IPOInsight is a fintech web application that helps retail investors analyze IPOs using financial data, subscription statistics, GMP trends and AI-generated insights. Rather than simply displaying IPO information, the platform provides decision support through scoring, comparison, risk analysis and personalized recommendations.

Problem Statement

IPO information is scattered across multiple websites. Beginners struggle to interpret financial ratios, compare IPOs and assess risks, leading to uninformed investment decisions.

Existing Solutions

Platforms such as Chittorgarh, Moneycontrol, NSE, BSE, Groww and Zerodha provide IPO information, subscription data and GMP. However, they largely present raw information and limited decision-support capabilities.

Proposed Solution

Build a centralized Java Spring Boot platform that aggregates IPO data, analyzes company fundamentals, calculates an IPO Score, compares offerings, explains financial metrics and generates AI-powered investment summaries.

Key Features

- Secure Login (JWT)
- IPO Dashboard
- Company Fundamentals
- Financial Ratio Analysis
- IPO Comparison
- IPO Score (0-100)
- Risk Analyzer
- AI Summary
- Personalized Recommendation Engine
- Watchlist & Notifications
- Historical IPO Analytics
- Learning Hub

Technology Stack

Frontend: React + Tailwind CSS

Backend: Java 21, Spring Boot, Spring Security, JWT, Hibernate/JPA

Database: PostgreSQL

Tools: Maven, Git, Postman, Swagger

Charts: Chart.js

System Architecture

React Frontend -> Spring Boot REST APIs -> Business Logic -> PostgreSQL + External Financial APIs -> Analytics Engine -> Dashboard

Suggested APIs

Finnhub (IPO Calendar, Fundamentals), Financial Modeling Prep (Financial Statements), Alpha Vantage (Market Data), NewsAPI/GNews (Financial News), OpenAI/Gemini (AI Summaries).

Future Enhancements

ML-based listing gain prediction, sentiment analysis, portfolio tracking, mobile app, chatbot, Docker deployment.

Conclusion

IPOInsight is positioned as a financial decision-support platform rather than an IPO information website. It demonstrates full-stack development, API integration, authentication, analytics and AI-assisted insights, making it suitable as a final-year Java project.